Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.Tech CSE 5th semester.
Name of the Paper: Computer Based Numerical & Statistical Techniques.
Paper Code: TMA-502
Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.
(iii) Please specify COs against each question.

Q1. (10 Marks)

a. (i). If $\Delta x = 0.005$, $\Delta y = 0.001$ be the absolute errors in $x = 2.11$ and $y = 4.15$, find the relative and percentage error in the computation of $x + y$.
(ii) Let $x = 0.458529 * 10^{-2}$. Find the absolute, relative & percentage error if x is rounded-off to 3 decimal places. (CO1)

OR

b. Obtain the root of the equation $x^3 + x^2 + x + 7 = 0$ correct to 3 decimal places by bisection method. (CO1)

Q2. (10 Marks)

a. Find the real root of the equation $2x = \cos x + 3$ using Newton Raphson method correct to 3 decimal places. (CO1)

OR

b. Find the real root of the equation $2x - \log_e x = 6$ using Regula falsi method correct to 3 decimal places. (CO1)

Q3. (10 Marks)

a. Solve the following system of linear equations by Gauss Jordan Method: $x + y + 2z = 4$; $3x + y - 3z = -4$; $2x - 3y - 5z = -5$. (CO1)

OR

b. Solve the following system of linear equations by Gauss Seidal Method: $10x + y + z = 9$; $2x + 10y + z = 13$; $2x + 2y + 10z = 14$. (CO1)

Q4. (10 Marks)

a. In an examination, the number of candidates who secured marks between certain limits were as follows:

Marks:	0-19	20-39	40-59	60-79	80-99
No. of Students:	41	62	65	50	17

Estimate the number of candidates getting marks less than 70. (CO2)

OR

b. Compute the value of Y when X = 0.6538 by using Gauss Forward formula, given that

X:	0.62	0.63	0.64	0.65	0.66	0.67	0.68
Y:	0.619411	0.627046	0.634857	0.642029	0.649376	0.656628	0.663782

(CO2)

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Q5. (10 Marks)

a. Use Newton's Divided Difference formula to find the value of F(5.60275) from the following table:

X:	5.600	5.602	5.605	5.607	5.608
F(X):	0.775566	0.776827	0.778712	0.779966	0.780591

(CO2)

OR

b. Use Lagranges formula of interpolation to find the value of Y(5.5), given that:

X:	0	3	4	6
Y:	1	19	49	181

(CO2)

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